Marcel Harrer, M.Sc.

Senior AI Engineer · LLM Systems, RAG & Multi-Agent Architectures
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Summary

AI engineer focused on production LLM and agent systems: RAG, multi-agent orchestration, and the infrastructure to run them reliably. Currently building a multi-agent system that automates email processing for 50 government departments. Applied-mathematics background, with neural-network approximation research nearing publication.

Experience

Senior AI Engineer, Fivesquare *Remote · Mar 2026 – Present*

- Built and shipped a multi-agent system that automatically processes incoming government email across 50 departments: around 10 million emails a year, previously handled by hand, each attached to the correct case file among thousands with 90% accuracy.
- Architected it as a shared orchestrator routing to a dedicated agent per department, each learning new skills from experience (an agentic context engineering approach) rather than needing constant retraining.
- Own the system end to end in production on the core AI team: design, deployment, and operation.

AI Engineer (Freelance), Independent *Remote · Mar 2025 – Dec 2025*

- Built a Text-to-SQL agent (LangGraph) that lets non-technical staff pull data from PostgreSQL databases in plain English, removing their dependence on engineers for routine reports.
- Designed the modular tool interfaces the agent uses to query databases, reason over structured data, and trigger email workflows, so it handles multi-step requests end to end instead of one-off queries.
- Tuned agent orchestration and prompt strategies so the system returned reliable, repeatable results in production, not just a working demo.

AI Research Engineer, ImageTwin *Jan 2024 – Dec 2025*

Part-time in 2025, concurrent with the freelance role above.

- Built computer-vision models that flag manipulated and duplicated images in scientific figures by matching them against a 100-million-image dataset, surfacing integrity problems in around 100 papers a year that are nearly impossible to spot by eye.
- Raised classification accuracy for the most common bioscience image types to 98%; implemented keypoint matching (OpenCV SIFT/SURF) and trained LightGlue models to match reused or altered regions.
- Co-built an end-to-end pipeline to detect AI-generated images, from labeling training data through automated evaluation, as newer fakes began slipping past existing checks.

AI Engineer, Financial Market Authority (FMA) *Vienna, Austria · Dec 2022 – Dec 2023*

- Built an AI search assistant (RAG with FAISS and transformer embeddings) that gave regulators instant, source-backed answers from thousands of German-language regulatory documents.
- Made large regulatory archives searchable by meaning rather than exact wording (SBERT similarity), so staff could surface relevant rules they would otherwise miss.
- Turned complex Austrian insurance-sector data into clear R dashboards for non-technical decision-makers.

Research & Selected Projects

B-Spline Networks and a Universal Approximation Theorem *(in preparation, with Prof. Arnold Neumaier)*
Introduced "spline networks", single-hidden-layer networks whose fixed activations are replaced by adaptive cubic splines over random ridge directions, and proved a quantitative universal approximation theorem bounding the expected L^2 error by $O(1/m + h^8)$. Training reduces to a sparse linear least-squares problem solved with conjugate gradients in Python.

Master's Thesis: Accelerating Gradient Descent via L-Smoothness
Adaptive optimization algorithm that uses L-smoothness to dynamically tune step sizes, with provable convergence guarantees and improved speed over fixed-step methods.

Education

M.Sc. in Applied Mathematics, University of Vienna, passed with honors *Oct 2020 – Nov 2023*
B.Sc. in Mathematics, University of Vienna, passed with honors *Oct 2016 – Sep 2020*

Technical Skills

Languages: Python, SQL
LLMs & Agents: RAG, multi-agent orchestration, LangGraph, LangChain, MCP (Model Context Protocol), prompt engineering, Hugging Face Transformers, vLLM, Ollama, AWS Bedrock
ML & Computer Vision: PyTorch, TensorFlow, OpenCV (SIFT/SURF), LightGlue, time-series forecasting
Retrieval & Data: FAISS, Qdrant, pgvector, semantic search, PostgreSQL
MLOps & Infrastructure: FastAPI, Docker, Kubernetes, CI/CD, AWS (SageMaker, Lambda, S3, EC2), Weights & Biases, Langfuse
Spoken Languages: German (native), English (bilingual, C2), Spanish (advanced, C1)